When Borders Become Distant:

Accessibility and Economic Activity along the Colombia–Venezuela Border

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Abstract

This paper examines how changes in international border accessibility relate to local economic activity along the Colombia–Venezuela border. Implementing a spatial-accessibility framework, changes in shortest-path road-network distance to operational international crossings are combined with quarterly VIIRS nighttime-light observations. The results show a negative relationship between deteriorating border access and local economic activity after 2019, but the estimated effect becomes smaller and statistically insignificant when increasingly localized time-varying shocks are controlled for. Event-study estimates further reveal differential pre-treatment dynamics, limiting a causal interpretation of the baseline relationship. The findings nevertheless demonstrate that border restrictions substantially altered effective economic geography and illustrate the value of combining GIS and satellite data to study economic activity in data-scarce environments.

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1 Introduction

International borders are not only political boundaries. They also shape access to markets, transportation networks, employment opportunities, and trade. For communities located in border regions, changes in the ability to cross an international boundary can generate substantial changes in effective economic geography by altering their access to nearby markets (Redding and Sturm, 2008). A border closure, or reopening, may leave the physical location of a municipality unchanged while sharply affecting the distance that firms and households must travel to reach an operational crossing. These episodes provide a particularly useful setting in which to study the relationship between transport accessibility and local economic activity.

This paper studies this relationship along the border between Colombia and Venezuela. The two countries share a border characterized by deep historical, social, and economic connections. In several parts of the border, economic activity has historically developed across rather than independently on each side of the international boundary. Bilateral political tensions and repeated restrictions on border crossings have therefore had consequences extending beyond international trade flows, affecting local commercial activity, employment, mobility, and informal economic exchange (Rodriguez-Hernandez, 2009; Bustamante, 2016; Hernandez, 2024). The deterioration of diplomatic relations between Colombia and Venezuela culminated in a major disruption of formal border connectivity in 2019.

The spatial heterogeneity in municipalities exposure to changes in border accessibility provides the basis for the empirical strategy of this paper, following the methodology developed by Behrens (2024). Rather than classifying municipalities simply as treated or untreated according to whether they are located directly adjacent to the border, a continuous measure of treatment intensity based on changes in road-network accessibility is constructed. This approach captures the fact that the economic importance of a border restriction depends not only on geographic proximity to the international boundary, but also on the configuration of the transportation network and the availability of alternative crossings.

The analysis combines several spatial data sources to construct a municipality-level panel of border accessibility and local economic activity. Administrative boundaries for Colombia and Venezuela are combined with georeferenced international border crossings and OpenStreetMap road-network data to measure each municipality's shortest road distance to an operational crossing under different border regimes. These accessibility measures are then matched with quarterly satellite observations of nighttime luminosity, which provide the main measure of local economic activity. The resulting panel links quarterly nighttime-light intensity to spatial variation in the accessibility shock experienced by each municipality.

The empirical strategy begins with a conventional two-way fixed-effects specification including municipality and quarter fixed effects. Progressively more demanding controls for spatially heterogeneous shocks are then introduced, including country-by-quarter, crossing-corridor-by-quarter, and state/department-by-quarter fixed effects. The average post-treatment estimates are complemented by event-study specifications that examine the evolution of the relationship between treatment intensity and nighttime lights before and after the 2019 border disruption.

The conventional two-way fixed-effects specification indicates a negative relationship between deteriorating border access and nighttime luminosity. However, the estimate becomes smaller under increasingly localized time controls, while event-study diagnostics reveal differential pre-treatment dynamics. The evidence therefore suggests that border accessibility is economically relevant, but does not support interpreting the baseline relationship as a clean causal effect of the 2019 disruption.

The findings highlight both the potential and the limitations of using spatial policy shocks for causal inference. The road-network analysis shows that changes in border policy produced substantial geographic variation in effective access to international crossings. Yet the event-study results also show that municipalities receiving different levels of exposure were already following different economic trajectories before the 2019 disruption. Rather than treating this evidence as incidental, it is incorporated into the interpretation of the results. The evidence is therefore consistent with an economically meaningful relationship between border accessibility and local economic activity, but it does not support a strong interpretation of the baseline estimate as a clean causal effect of the closure.

2 Motivation

The Colombia–Venezuela border provides a particularly relevant setting for studying the economic consequences of changes in cross-border accessibility. Unlike borders that primarily separate otherwise independent regional economies, large sections of the Colombian–Venezuelan frontier have historically functioned as integrated economic and social spaces. This is especially evident along the Norte de Santander–Táchira corridor, where commercial activity, transportation, employment, migration, and household relationships have developed across the international boundary. The economic interdependence of these regions has led the Colombia–Venezuela frontier to be described as one of the most active borders in Latin America (Pabón León et al., 2016). Political and economic disturbances in either country can therefore have direct consequences for communities on the opposite side of the border.

Historically, Venezuela represented a major commercial partner for Colombia and an especially important market for the Colombian border departments. The Norte de Santander–Táchira corridor concentrated a substantial share of the economic interaction between the two countries, while Venezuela accounted for a large proportion of exports from Norte de Santander during periods of stronger bilateral integration (Pabón León et al., 2016; Hernandez, 2024). Consequently, the economic relevance of the border cannot be understood exclusively through national bilateral trade statistics. For municipalities within the border region, access to an international crossing determines connections to nearby cities, consumers, suppliers, employment opportunities, and informal markets located on the opposite side of the boundary.

This dependence on cross-border accessibility became particularly important as political relations between the two countries deteriorated. In August 2015, the Venezuelan government unilaterally closed major crossings along the Táchira–Norte de Santander border, beginning a period in which formal mobility across the frontier became increasingly dependent on political decisions and changing crossing restrictions (Sayago Rojas, 2016; Bustamante, 2019). Subsequent restrictions, diplomatic tensions, and disruptions to formal crossings altered the geography through which border communities could interact.

A second motivation arises from the difficulty of measuring economic activity during the Venezuelan economic crisis. Conventional empirical analysis would ideally rely on comparable and regularly reported measures of output, employment, income, or commercial activity at a sufficiently disaggregated spatial and temporal level on both sides of the border. Such information is particularly difficult to obtain for Venezuela. During the second half of the 2010s, the availability and regular publication of official Venezuelan macroeconomic statistics deteriorated substantially (Rodríguez, 2020). The International Monetary Fund notes that the assessment of economic developments in Venezuela is complicated by limited reported statistics, incomplete metadata, and difficulties reconciling available indicators with economic developments. For most macroeconomic indicators between 2018 and 2022, the IMF therefore relies on staff estimates and explicitly cautions that the scarcity of reported data creates substantial uncertainty (International Monetary Fund, 2024). More recently, official GDP publication by the *Banco Central de Venezuela* has remained unavailable beyond the first quarter of 2019, further motivating the use of alternative sources for measuring economic activity (Fund, 2026).

The statistical problem becomes even more severe at the local level. Even when national estimates are available, they provide little information about how economic conditions evolved across individual municipalities along the border. This is particularly problematic for evaluating border restrictions because their economic effects are inherently spatial. Similarly, using economic information from only the Colombian side would provide an incomplete representation of a border economy whose defining characteristic is precisely the interaction between communities located in two different countries.

These limitations make remotely sensed and geographical data especially valuable. Satellite-recorded nighttime luminosity provides a consistently measured indicator available on both sides of the international border, independently of national statistical systems. The use of nighttime lights as a proxy for economic activity is well established in the economic literature, particularly in settings where conventional statistics are incomplete or unreliable (Henderson et al., 2012).

3 Institutional Background

3.1 The Colombia–Venezuela border

Colombia and Venezuela share an extensive international border stretching from the Caribbean coast in the north to the Orinoco basin in the south. Despite its length, formal cross-border movements are concentrated at a relatively small number of international crossings. This creates a transportation geography in which access to the neighbouring country depends not simply on proximity to the international boundary, but on the location of operational crossings and the road network connecting municipalities to them (Pabón León et al., 2016; Hernandez, 2024).

Figure 1 illustrates this geography. The formal crossings considered in the analysis are distributed unevenly along the frontier, with an important concentration around the Norte de Santander–Táchira corridor. This corridor connects the metropolitan area of Cúcuta and Villa del Rosario in Colombia with San Antonio del Táchira, Ureña, and the wider urban system of Táchira in Venezuela. Its principal infrastructure includes the Simón Bolívar and Francisco de Paula Santander international bridges. Farther north, the Paraguachón crossing connects La Guajira with the Venezuelan state of Zulia, while the José Antonio Páez bridge connects Arauca with the Venezuelan state of Apure. The importance of these formal corridors reflects the broader pattern of economic and social interaction that has historically characterized the frontier (Pabón León et al., 2016; Hernandez, 2024).

3.2 Border restrictions and the 2019 disruption

Although bilateral tensions intensified considerably toward the end of the 2010s, restrictions on the Colombia–Venezuela border preceded the main event studied in this paper. A major episode occurred in August 2015, when the Venezuelan government closed crossings in the state of Táchira, including the Simón Bolívar and Francisco de Paula Santander bridges. The measure formed part of a broader political

Figure 1: Colombia–Venezuela border, analysis area, and international crossings

Notes: The main map displays the municipalities included in the geographical analysis and the road network used to calculate accessibility to the international border crossings. The inset locates the analysis area within Colombia and Venezuela. C01 denotes the Simón Bolívar International Bridge; C02 the Francisco de Paula Santander International Bridge; C03 the Atanasio Girardot International Bridge; C04 the Paraguachón crossing; and C05 the José Antonio Páez International Bridge. Road-network information is derived from OpenStreetMap.

and security crisis between the two countries and substantially disrupted established patterns of mobility and exchange along the frontier (Sayago Rojas, 2016; Bustamante, 2019). The border subsequently underwent a process of partial reopening. In August 2016, the governments of Colombia and Venezuela announced a gradual, controlled reopening beginning on 13 August, initially allowing pedestrian passage through five authorized border points (Ministerio de Relaciones Exteriores de Colombia, 2016). The reopening was therefore partial and did not imply an immediate restoration of the previous commercial border regime.

The institutional environment nevertheless remained unstable. Formal cross-border activity remained sensitive to diplomatic tensions, security measures, and changes in the conditions under which individual crossings could operate. These developments culminated in February 2019 during the confrontation surrounding the attempted delivery of humanitarian aid from Colombia into Venezuela. The Colombian government established humanitarian distribution operations in Cúcuta and introduced special restrictions on movement through official migration-control posts in Norte de Santander during the planned operation on 23 February (Presidencia de la República de Colombia, 2019; Migración Colombia, 2019). The episode coincided with a sharp deterioration in bilateral political relations and a renewed disruption of the formal border regime.

For the purposes of this paper, the economically relevant feature of the 2019 episode is the interruption of formal commercial connectivity through the Norte de Santander crossings. According to the Colombian Ministry of Foreign Affairs, February 2019 marked the last definitive closure of this section of the frontier to foreign-trade traffic before the restoration of commercial operations in 2022 (Ministerio de Relaciones Exteriores de Colombia, 2022). Merchandise that continued to move formally between Colombia and Venezuela during this period had to rely principally on the Paraguachón crossing in La Guajira or on maritime transport through Cartagena. Colombian authorities explicitly noted that this rerouting increased transport costs and travel times for firms that continued importing and exporting (Ministerio de Relaciones Exteriores de Colombia, 2022).

This distinction is central to the identification strategy. The empirical shock is not interpreted as the disappearance of all interaction across the international boundary. Consequently, the conditions governing passenger, pedestrian, informal, and formal commercial traffic did not necessarily coincide [1]. The analysis therefore focuses specifically on changes in access through the formal crossing network relevant for road-based commercial connectivity.

3.3 Reopening and the restoration of formal connectivity

The restoration of formal connectivity occurred gradually rather than through a single reversal of the 2019 restrictions. Border policy was further complicated by the COVID-19 pandemic. In March 2020, Colombia ordered the closure of authorized land and river crossings with Venezuela as part of its public-health measures, and restrictions on formal border movements continued through subsequent stages of the pandemic (Presidencia de la República de Colombia, 2020). The gradual normalization of migration controls began later, but this did not immediately imply the restoration of the commercial transportation network that had operated before 2019.

A more substantial institutional change followed the restoration of diplomatic relations between Colombia and Venezuela in 2022. On 26 September 2022, formal cargo traffic resumed through the Simón Bolívar and Francisco de Paula Santander international bridges in Norte de Santander. Colombian authorities described the event as the restoration of cargo movement through this section of the frontier after its interruption in February 2019 (Ministerio de Relaciones Exteriores de Colombia, 2022). The reopening therefore reversed an important component of the accessibility shock studied in the main analysis.

The process continued on 1 January 2023 with the entry into operation of the Atanasio Girardot International Bridge, previously known as the Tienditas bridge. The bridge connects Villa del Rosario in Norte de Santander with the municipality of Pedro María Ureña in Táchira and had been constructed before the deterioration in bilateral relations but had not entered normal international operation. Its opening in 2023 provided an additional route for international cargo and passenger transportation (Ministerio de Relaciones Exteriores de Colombia, 2023). Colombian government sources subsequently described the Simón Bolívar, Francisco de Paula Santander, and Atanasio Girardot crossings as part of the restored commercial and transportation integration between the two countries (Presidencia de la República de Colombia, 2023).

Overall, the institutional history demonstrates why border accessibility cannot be represented adequately by a simple indicator for proximity to the Colombia–Venezuela boundary. Formal connectivity changed through specific crossing points and at different moments in time, producing heterogeneous changes in effective road-network access across municipalities. The empirical analysis translates these institutional changes into alternative crossing regimes and calculates the corresponding shortest-path distances for each municipality. Appendix Table 2 summarizes the crossing-status changes used to construct these regimes, including the relevant dates, operational classifications, and supporting institutional evidence.

[1] Informal crossings, commonly referred to as *trochas*, continued to facilitate movements of people and goods in parts of the frontier, particularly around Cúcuta, San Antonio, and Ureña. Ethnographic research documents that these informal routes remained important during periods in which formal crossings were closed or heavily restricted (Hernandez, 2024).

3.4 Relation to the existing literature

This paper relates to three strands of literature. First, research on the Colombia–Venezuela frontier emphasizes the historically integrated character of the border economy and the importance of cross-border mobility, trade, and commercial activity for communities on both sides of the boundary (Pabón León et al., 2016; Bustamante, 2019; Hernandez, 2024). This literature also documents how political tensions and restrictions on formal crossings disrupted established patterns of mobility and exchange. The present analysis builds on this institutional evidence by measuring the spatial incidence of these restrictions across municipalities through changes in road-network accessibility.

Second, the paper is related to the broader literature on transportation, borders, and market access. Changes in transport connectivity can alter the effective economic distance between locations and consequently their access to markets even when their physical locations remain unchanged (Redding and Sturm, 2008; Donaldson and Hornbeck, 2016). Most directly, the empirical design implements the spatial-accessibility approach developed by Behrens (2024), who uses changes in access to international crossings to study the local consequences of border changes. The present study is not a replication of that analysis. Rather, it applies its central spatial logic to the Colombia–Venezuela frontier, constructing treatment from changes in shortest-path road-network distance to operational crossings under the institutional regimes described above.

Finally, the analysis relates to the literature using remotely sensed nighttime luminosity to measure economic activity where conventional statistics are unavailable or spatially incomplete. Nighttime lights provide a geographically consistent measure that can be observed independently of national statistical systems (Henderson et al., 2012; Zhang et al., 2020). This feature is especially valuable in the present setting given the limitations of official Venezuelan economic statistics and the difficulty of obtaining comparable municipality-level outcomes on both sides of the border. Taken together, these literatures motivate the combination of network-based accessibility and satellite-recorded luminosity used in the empirical analysis.

4 Conceptual framework

4.1 Border accessibility and local economic activity

The conceptual starting point of the analysis is that international borders affect local economic activity not only through their physical location, but also through the accessibility of the infrastructure that permits them to be crossed. A municipality may remain at the same geographic distance from an international boundary while experiencing a substantial change in its effective access to the neighbouring country if a nearby crossing becomes unavailable. In this sense, border restrictions can be interpreted as shocks to economic distance rather than changes in physical geography.

This distinction is particularly relevant along the Colombia–Venezuela border. As discussed in Section 3, formal interaction is concentrated at a limited number of crossing points connected to municipalities through the road network. The closure of a crossing can therefore force firms, households, and transporters to use a more distant alternative route. Municipalities that previously depended on the affected crossing experience a larger increase in effective distance than municipalities whose nearest operational crossing remains unchanged.

The economic mechanism underlying the analysis follows the broader idea of market access: transportation costs influence the set of consumers, suppliers, workers, and markets that can be reached from a particular location. Improvements in transport connectivity can increase market access and economic activity, whereas disruptions to transport networks can produce the opposite effect (Redding and Sturm, 2008; Donaldson and Hornbeck, 2016). International borders can amplify this mechanism because crossing costs and the location of border infrastructure determine where international economic interaction can occur.

4.2 Implementation of the Behrens framework

The empirical approach implements and adapts the spatial accessibility framework developed by Behrens (2024). It uses the underlying methodological insight, that changes in the availability of border crossings can generate heterogeneous changes in effective distance, and applies it to a different institutional setting, geographical context, and economic shock.

In the original application, Behrens (2024) exploits changes in border configuration and uses transportation-network distances to construct spatially heterogeneous exposure to changes in market access. The relevant treatment is therefore not defined solely by whether an observation is located on one side of a border or within an arbitrary geographic buffer. Instead, exposure depends on how the relevant institutional change modifies the transportation distance connecting a location to available border infrastructure.

Following the spatial-accessibility logic developed by Behrens (2024), exposure is allowed to vary continuously according to the deterioration in effective access to operational border crossings. The precise construction of this measure and its incorporation into the empirical specification are presented in Section 5.

A municipality for which the nearest relevant crossing remains available has a treatment intensity close to zero. By contrast, a municipality that loses access to its previously closest crossing and must travel substantially farther through the road network receives a larger positive value of T_i. The treatment is therefore continuous and captures the spatial heterogeneity generated by the change in formal border accessibility.

This construction has two advantages over simpler measures of border exposure. First, it incorporates the actual road network rather than relying on Euclidean distance. This matters because straight-line proximity does not necessarily correspond to transportation accessibility, particularly where road infrastructure is unevenly distributed. Second, the treatment incorporates the location and operational status of specific international crossings. Two municipalities located at similar distances from the international boundary may therefore receive very different treatment intensities if the 2019 disruption changes their feasible routes differently.

The relevant crossing regimes are reconstructed from the institutional history described in Section 3, while shortest-path distances are calculated using the road network connecting Colombian and Venezuelan municipalities to the formal international crossings included in the analysis. Economic activity is measured using quarterly nighttime luminosity, allowing the resulting accessibility shock to be related to a geographically consistent outcome on both sides of the border.

4.3 Empirical hypotheses

The first hypothesis follows directly from the transportation-cost and market-access mechanism. If access to the neighbouring country contributes to local economic activity, municipalities experiencing larger increases in road-network distance to an operational crossing should exhibit weaker economic performance following the disruption. Since nighttime luminosity is used as the observable proxy for local economic activity, the primary hypothesis is

> **H1:** Municipalities experiencing larger increases in road-network distance to an operational international crossing after the 2019 disruption experience a relative decline in nighttime-light intensity.

Under the treatment definition in Equation (2), this implies an expected negative relationship between the interaction of treatment intensity with the post-2019 period and nighttime luminosity.

A second implication concerns spatial heterogeneity. The economic relevance of the shock should depend on how strongly the disruption changes effective accessibility. Municipalities for which the nearest operational crossing remains unchanged should experience comparatively little treatment, whereas locations forced to make substantial road-network detours should be more strongly exposed. This gives the second hypothesis:

> **H2:** The economic response to the 2019 disruption is stronger in municipalities experiencing larger deterioration in effective border accessibility.

The institutional setting also suggests that effects need not be identical on the two sides of the border. Colombia and Venezuela entered the 2019 episode under substantially different macroeconomic and institutional conditions, while border municipalities differed in their dependence on cross-border markets and their available alternatives. Country-specific estimates are therefore useful for identifying heterogeneous responses, but no requirement is imposed that the effect be symmetric across countries.

Finally, the reopening of formal crossings beginning in September 2022 provides a complementary implication of the same mechanism. If deterioration in border accessibility reduces local market access, restoring previously unavailable routes should reduce effective transportation distances and may improve economic activity in municipalities that benefit most from the reopening. This motivates the complementary hypothesis

> **H3:** Municipalities experiencing larger improvements in border accessibility following the restoration of formal crossings exhibit a relative increase in nighttime-light intensity.

The reopening hypothesis is treated as a complementary robustness exercise rather than as the principal identification strategy. The period surrounding 2022 differs substantially from that surrounding the original 2019 disruption, including the effects of the COVID-19 pandemic and the broader normalization of diplomatic and commercial relations between Colombia and Venezuela. A reversal in the sign of the relationship is therefore informative about the proposed accessibility mechanism, but it should not be interpreted as a symmetric experimental reversal of the original shock.

5 Empirical methodology

The empirical strategy examines whether the deterioration in access to operational international crossings following the 2019 border disruption was associated with changes in local economic activity. Following the conceptual framework developed in Section 4, the analysis exploits cross-sectional variation in the extent to which the disruption altered each municipality's effective distance to the Colombian–Venezuelan border.

5.1 Measuring changes in border accessibility

Let d_{ic} denote the shortest road-network distance between municipality i and international crossing c. For a given border regime r, only a subset of crossings is operational. The effective border-access distance of municipality i under regime r is therefore defined as

$$d_{ir} = \min_{c \in C_r} \{d_{ic}\}, \tag{1}$$

where C_r denotes the set of crossings considered operational under regime r.

This definition allows the relevant crossing to differ across municipalities and border regimes. If the crossing initially providing the shortest route from a municipality becomes unavailable, the municipality is reassigned to the nearest crossing that remains operational. The resulting change in distance therefore captures the detour imposed by the disruption rather than simply the municipality's physical proximity to the international boundary.

Let d_i^{pre} denote effective border-access distance under the pre-disruption regime and d_i^{post} the corresponding distance under the border regime following the 2019 disruption. The principal measure of treatment intensity is

$$T_i = \log\left(d_i^{post}\right) - \log\left(d_i^{pre}\right). \tag{2}$$

A value of zero indicates that the disruption produces no change in the municipality's effective distance to an operational crossing. Positive values represent deteriorations in accessibility, with larger values corresponding to larger proportional increases in road-network distance.

The logarithmic difference captures proportional rather than absolute changes in accessibility. This distinction is relevant because an additional kilometre of travel need not have the same economic significance for a municipality initially located close to a crossing as for one already located far from it [2].

5.2 Baseline empirical specification

The baseline empirical specification relates the municipality-specific accessibility shock to changes in nighttime luminosity following the 2019 border disruption. Let $Post_t$ be an indicator equal to one during the post-disruption period and zero beforehand. The principal treatment variable is the interaction

$$Exposure_{it} = T_i \times Post_t. \tag{3}$$

The conventional two-way fixed-effects specification is

[2] As a robustness exercise, the analysis also considers the absolute change $\Delta d_i = d_i^{post} - d_i^{pre}$.

$$y_{it} = \alpha_i + \lambda_t + \beta\,(T_i \times Post_t) + \varepsilon_{it}, \tag{4}$$

where y_{it} denotes nighttime-light intensity in municipality i and quarter t, α_i represents municipality fixed effects, λ_t represents quarter fixed effects, and ε_{it} is the error term.

Municipality fixed effects absorb time-invariant differences across locations, including persistent differences in geography, infrastructure, settlement patterns, and baseline economic activity. Quarter fixed effects absorb shocks common to all municipalities in a particular period in time.

The coefficient of interest is β. Under the hypothesis developed in Section 4, a deterioration in border accessibility should reduce local economic activity, implying $\beta < 0$.

Because T_i is time invariant, its level is absorbed by the municipality fixed effects. Similarly, $Post_t$ is absorbed by the quarter fixed effects. The coefficient β is therefore identified from differences in the evolution of nighttime luminosity after the disruption between municipalities experiencing different changes in effective border accessibility.

Standard errors are clustered at the municipality level throughout the main analysis. This allows the regression residuals to be arbitrarily correlated over time within a municipality.

5.3 Spatially heterogeneous time shocks

The conventional two-way fixed-effects model assumes that, after accounting for municipality fixed effects, the remaining time shocks are common across the study area. This assumption may be restrictive in the Colombia–Venezuela setting. Municipalities on opposite sides of the border were exposed to substantially different national economic and political conditions, while economic trajectories may also have differed across border corridors and first-level administrative regions.

The empirical strategy therefore progressively replaces common quarter fixed effects with more geographically disaggregated time effects.

The first extension allows each country to experience an independent shock in every quarter:

$$y_{it} = \alpha_i + \lambda_{c(i)t} + \beta\,(T_i \times Post_t) + \varepsilon_{it}, \tag{5}$$

where $\lambda_{c(i)t}$ denotes country-by-quarter fixed effects. Identification in this specification comes from comparing municipalities with different treatment intensities within the same country and quarter.

This distinction is especially important because the empirical period overlaps with the Venezuelan economic crisis. Country-by-quarter fixed effects absorb aggregate national shocks that may otherwise be correlated with both nighttime luminosity and the timing of the border disruption.

A more demanding specification controls for shocks shared by municipalities associated with the same pre-disruption border-crossing corridor:

$$y_{it} = \alpha_i + \lambda_{c(i)t} + \gamma_{g(i)t} + \beta\,(T_i \times Post_t) + \varepsilon_{it}, \tag{6}$$

where $\gamma_{g(i)t}$ represents crossing-corridor-by-quarter fixed effects, with municipalities assigned to the crossing providing their nearest operational route prior to the disruption. This specification compares municipalities exposed to different changes in accessibility while allowing each initial crossing corridor to follow its own time path.

Finally, the preferred local-control specification introduces state/department-by-quarter fixed effects:

$$y_{it} = \alpha_i + \delta_{s(i)t} + \beta\,(T_i \times Post_t) + \varepsilon_{it}, \tag{7}$$

where $\delta_{s(i)t}$ denotes a separate fixed effect for every state/department and quarter.

This specification absorbs arbitrary time-varying shocks common to municipalities within the same first-level administrative region. Consequently, β is identified only through differences in treatment intensity between municipalities located within the same state or department in the same quarter. This provides a substantially more demanding comparison than the conventional two-way fixed-effects specification.

5.4 Country-specific effects

The effect of changing border accessibility need not be symmetric across Colombia and Venezuela. The two countries entered the border disruption under different macroeconomic and institutional conditions, and municipalities on each side may differ in their dependence on cross-border economic activity.

To examine this heterogeneity, the treatment interaction is allowed to vary by country:

$$\begin{aligned} y_{it} = {} & \alpha_i + \lambda_{c(i)t} \\ & + \beta_C\,(T_i \times Post_t \times Colombia_i) \\ & + \beta_V\,(T_i \times Post_t \times Venezuela_i) + \varepsilon_{it}, \end{aligned} \tag{8}$$

where β_C and β_V capture the association between the accessibility shock and nighttime luminosity separately for Colombian and Venezuelan municipalities.

These estimates are interpreted as tests of spatial heterogeneity rather than as requiring the treatment effect to have opposite signs across the two countries. As discussed in Section 4, the conceptual framework predicts that accessibility matters for local activity but does not impose an ex ante requirement that its magnitude be identical on both sides of the border.

5.5 Dynamic effects and pre-treatment trends

The identifying interpretation of Equation (4) requires that municipalities with different levels of treatment intensity would not have followed systematically different economic trajectories in the absence of the border disruption. Because treatment intensity is spatially structured, this assumption deserves particular attention.

An event-study specification is therefore used to examine the relationship between subsequent treatment intensity and nighttime luminosity before and after the disruption:

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k\,[T_i \times \mathbb{1}\,(EventTime_t = k)] + \varepsilon_{it}, \tag{9}$$

where k indexes quarters relative to the beginning of the 2019 disruption and the quarter immediately preceding treatment, $k = -1$, is omitted as the reference period.

The coefficients β_k describe how differences in nighttime luminosity associated with treatment intensity evolve relative to the omitted quarter. Post-treatment coefficients provide evidence on the timing and persistence of the relationship between accessibility and luminosity. The pre-treatment coefficients are particularly important because they reveal whether municipalities receiving different subsequent treatment intensities were already following different trajectories before the disruption.

The event-study analysis is estimated under alternative fixed-effects structures, including country-by-quarter, crossing-corridor-by-quarter, and state/department-by-quarter effects. Joint Wald tests of the pre-treatment coefficients complement the graphical evidence.

These tests are treated as identification diagnostics rather than as a criterion that must mechanically be satisfied. Statistically significant pre-treatment differences indicate that the identifying assumptions required for a strict causal interpretation are less credible and therefore directly inform the interpretation of the post-treatment estimates.

5.6 Reopening and complementary evidence

The restoration of formal border connectivity beginning in 2022 provides a complementary test of the accessibility mechanism. The same network framework can be used to identify municipalities whose effective access improves when previously unavailable crossings become operational again.

If deteriorating accessibility reduces economic activity, the restoration of border access should, all else equal, produce an effect in the opposite direction. A reopening-exposure measure is therefore interacted with the post-reopening period and related to nighttime luminosity using the same general fixed-effects framework.

The reopening analysis is not interpreted as a symmetric reversal experiment. The economic and institutional environment in 2022 differed substantially from that prevailing in 2019, including the effects of the COVID-19 pandemic and changes in diplomatic relations between Colombia and Venezuela. Instead, the reopening exercise provides complementary evidence on whether improvements in network accessibility are associated with changes in local economic activity in the direction predicted by the conceptual mechanism.

6 Data

The empirical implementation combines four main sources of spatial information: administrative boundaries, the road network, international border crossings and their operational status, and nighttime luminosity. Together, these sources produce a municipality-quarter panel linking local economic activity to changes in effective access to the Colombia–Venezuela border. This section describes the underlying data, the construction of the accessibility and treatment measures, and the resulting estimation sample.

6.1 Administrative boundaries and study area

Administrative boundaries for Colombia and Venezuela are obtained from the Database of Global Administrative Areas (GADM). The analysis is conducted at the second administrative level, corresponding approximately to Colombian municipalities and Venezuelan *municipios*. Using a common international source provides a harmonized spatial framework for both countries and avoids combining administrative datasets constructed under different national statistical systems.

The geographic analysis focuses on municipalities located within 100 km of the Colombia–Venezuela international boundary. The 100 km buffer defines the main border study area and restricts the analysis to municipalities for which changes in cross-border accessibility are most economically relevant. All spatial layers are projected into a common coordinate reference system before distance and network operations are performed. For each municipality, a representative point guaranteed to lie within its administrative polygon is constructed and used as the origin of the road-network accessibility calculations.

The study area contains 235 municipalities. Of these, 192 are connected through the road network to the component containing the international crossings considered in the analysis. The remaining 43 municipalities belong to disconnected network components and therefore have no well-defined road-network distance to those crossings. Rather than assigning an artificial Euclidean or imputed distance, these municipalities are excluded from the baseline estimation sample.

6.2 Road network and international crossings

Road-network information is obtained from OpenStreetMap (OSM) extracts for Colombia and Venezuela. The network retains road classes potentially relevant for motorized accessibility, including motorway, trunk, primary, secondary, and tertiary roads and their associated link roads, together with unclassified, residential, living-street, service, and track roads. The geometries are restricted to the study area, validated, and processed into a routable network using the `dodgr` motorcar weighting profile.

Municipality representative points and international crossings are matched to their nearest network nodes, and shortest-path distances between every municipality and crossing are calculated using `cppRouting`. The resulting municipality–crossing distance matrix provides the basis for the treatment measure.

The analysis considers five principal formal crossing corridors: Puente Internacional Simón Bolívar, Puente Internacional Francisco de Paula Santander, Puente Internacional Atanasio Girardot, Paraguachón, and Puente Internacional José Antonio Páez. Bridge crossings are represented by their midpoint, while Paraguachón is represented directly by its land-crossing location. Figure 1 shows the crossing locations and road network within the study area.

The operational status of each crossing is reconstructed from historical and institutional sources describing changes in formal border access. A physical crossing is therefore not automatically considered available: its status depends on whether it permitted the type of formal road access relevant to the analysis during a given period.

The chronology of crossing-status changes and the sources used to classify each crossing are reported in Appendix Table 2. As a network-quality check, the crossings are located within approximately 6 to 139 metres of their assigned routing nodes. Network components are also retained explicitly, allowing municipalities without a valid route to the crossing network to be identified rather than assigned artificial distances.

6.3 Treatment intensity

The road-network distances and crossing-status information are combined to operationalize the continuous treatment measure defined in Section 5.

This construction implements the broader spatial-accessibility logic developed by Behrens (2024), adapted to the institutional and transportation characteristics of the Colombia–Venezuela border [3].

Among the 192 municipalities in the baseline sample, mean network distance increases from approximately 168 km before the disruption to 513 km under the closure regime. The average absolute increase

[3] Treatment intensity equals zero when the change in border regime does not alter a municipality's effective distance to an operational crossing, while larger positive values indicate a greater deterioration in border access.

is approximately 345 km. Treatment intensity has a mean and median of 1.20 and ranges from zero to approximately 4.60.

6.4 Nighttime luminosity

Local economic activity is measured using nighttime luminosity from the Visible Infrared Imaging Radiometer Suite (VIIRS). Specifically, the analysis uses the NASA Black Marble VNP46A3 product, which provides monthly Level-3 nighttime-light composites at a spatial resolution of 15 arc-seconds (Román et al., 2018; NASA VIIRS Land Science Investigator-led Processing System, 2025).[4]

The original monthly observations are converted into quarterly measures by taking the arithmetic mean of the available monthly composites for each raster cell within a calendar quarter. The resulting quarterly rasters, beginning in 2012Q2, are cropped and masked to the study area. Mean nighttime luminosity is then calculated for each municipality from the raster cells intersecting its administrative boundary, producing a consistently defined municipality-quarter outcome [5].

Let Light_{it} denote mean nighttime luminosity in municipality i and quarter t. Because luminosity is strongly right-skewed and includes observations equal or close to zero, the principal dependent variable is

$$y_{it} = \log\left(1 + \text{Light}_{it}\right). \tag{10}$$

The $\log(1 + x)$ transformation reduces the influence of highly luminous observations while retaining municipality-quarter observations with zero measured luminosity. The untransformed measure is retained as an alternative outcome in the robustness analysis.

6.5 Estimation sample

Combining the accessibility measures with quarterly nighttime-light observations produces the panel used in the empirical analysis. The baseline estimation sample contains 192 municipalities with valid road-network access to the international crossing network, of which 132 are located in Colombia and 60 in Venezuela. The baseline estimation window contains 21 quarters, yielding 4,032 municipality-quarter observations. There are no missing nighttime-light observations within this sample.

The descriptive statistics from Appendix B illustrate the spatial importance of the border-access shock. Municipalities in the baseline sample are located, on average, only 57 km from the international boundary, yet their mean road-network distance to an operational crossing increases from approximately 168 km before the disruption to 513 km under the closure regime. The average increase of approximately 345 km demonstrates why proximity to the international boundary alone provides an incomplete measure of exposure: access depends on both the location of operational crossings and the road network through which they can be reached.

Nighttime luminosity is similarly heterogeneous. Mean luminosity in the baseline panel is 0.727, compared with a median of 0.115 and a maximum of approximately 41.9, illustrating the strongly

[4] The data are accessed programmatically through the `blackmarbler` package. VNP46A3 is constructed from daily atmospherically and lunar-BRDF-corrected nighttime-light observations. The analysis uses the snow-free near-nadir composite radiance variable provided by the product.

[5] As discussed in Section 2, satellite observations are particularly useful in this setting because they provide a consistently measured indicator across both countries independently of their national statistical systems. The same product and processing procedure are therefore applied to Colombian and Venezuelan municipalities.

right-skewed distribution that motivates the logarithmic transformation used in the main specifications.

Taken together, the dataset links the three dimensions required by the empirical strategy: the spatial distribution of economic activity, the transportation network connecting municipalities to the border, and the institutional availability of individual international crossings over time.

7 Results

This section presents the empirical relationship between changes in border accessibility and municipality-level nighttime luminosity. The analysis proceeds from the conventional two-way fixed-effects specification to models that absorb increasingly localized time-varying shocks. This progression is important because municipalities experiencing larger changes in access to international crossings may also differ systematically in their underlying economic trajectories. The results therefore focus not only on the magnitude and statistical significance of the estimated coefficients, but also on their sensitivity to the geographic level at which common shocks are controlled for.

7.1 Baseline relationship between border access and economic activity

Table 1 reports the main estimates. The first specification includes municipality and quarter fixed effects and corresponds to the conventional two-way fixed-effects model.

The estimated coefficient on the interaction between treatment intensity and the post-2019 indicator is −0.0168, and statistically significant at the 5 percent level. The negative coefficient indicates that municipalities experiencing larger proportional increases in road-network distance to an operational international crossing also experienced comparatively weaker nighttime-light outcomes after 2019.

Table 1: Border Accessibility and Nighttime Luminosity

	(1) TWFE	(2) Country × Quarter	(3) Crossing × Quarter	(4) State/Dept. × Quarter
Treatment intensity × Post	-0.017^{**}	−0.010	−0.006	−0.006
	(0.008)	(0.007)	(0.010)	(0.011)
Municipality FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	No	No	No
Country × Quarter FE	No	Yes	Yes	No
Crossing corridor × Quarter FE	No	No	Yes	No
State/Department × Quarter FE	No	No	No	Yes
Observations	4,032	4,032	4,032	4,032
R^2	0.982	0.983	0.984	0.986
Adjusted R^2	0.981	0.982	0.983	0.984
Within R^2	0.011	0.005	0.001	0.001
Adjusted within R^2	0.011	0.004	0.001	0.000

Notes: The dependent variable is log(1 + nighttime lights). Treatment intensity is defined as $\log(d_i^{post}) - \log(d_i^{pre})$, where distance is the shortest road-network distance to the nearest operational international crossing. Post equals one during the clean closure period from 2019Q2 to 2022Q2; 2019Q1 is excluded as a transition quarter. Standard errors in parentheses are clustered at the municipality level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

This result is consistent with the mechanism developed in Section 4: deterioration in access to an international crossing may increase transportation costs and reduce access to markets, commercial activity, employment opportunities, and other forms of cross-border interaction. The baseline estimate therefore provides initial evidence of a negative relationship between the deterioration of border access and local economic activity.

The identifying assumption underlying a causal interpretation of this relationship, however, is demanding. Quarter fixed effects absorb shocks common to all municipalities, but they do not account for shocks that evolve differently across regions. This distinction is particularly relevant in the Colombia–Venezuela setting, where the two countries experienced substantially different economic and political conditions during the period under study. Furthermore, economic trajectories may differ even within countries, especially across states and departments located in different parts of the border region.

7.2 Accounting for spatially heterogeneous shocks

The second specification replaces common quarter fixed effects with country-by-quarter fixed effects. This allows Colombia and Venezuela to follow completely different time paths and therefore absorbs country-level shocks that vary over time. Under this specification, the estimated coefficient decreases in magnitude from -0.0168 to -0.0104. The estimate remains negative but is no longer statistically distinguishable from zero at conventional significance levels.

The attenuation of the coefficient is economically informative. It is consistent with part of the negative relationship observed in the conventional two-way fixed-effects model reflecting differences in economic trajectories between Colombia and Venezuela rather than variation in border accessibility alone. This is especially relevant given the severe macroeconomic contraction experienced by Venezuela during the period analysed.

The third specification further introduces crossing-corridor-by-quarter fixed effects, in addition to country-by-quarter fixed effects. Municipalities are thereby compared with other municipalities associated with the same pre-disruption international crossing and observed during the same quarter. The estimated coefficient declines further to -0.0062, and remains statistically insignificant.

Finally, the most geographically demanding main specification includes state/department-by-quarter fixed effects. Identification therefore comes from differences in treatment intensity among municipalities located within the same Colombian department or Venezuelan state and observed during the same quarter. The resulting coefficient is -0.0057, and is not statistically distinguishable from zero.

The pattern constitutes an important result of the analysis. The sign of the coefficient remains negative across all four specifications, but its magnitude is not stable once increasingly local time-varying shocks are absorbed. The evidence therefore does not support interpreting the statistically significant two-way fixed-effects coefficient as an isolated causal effect of the border-access shock.[6]

At the same time, the attenuation should not be interpreted as evidence that border accessibility is economically irrelevant. Rather, it indicates that separating the effect of accessibility from contemporaneous local economic conditions is difficult in this setting. The more demanding specifications compare municipalities within substantially narrower geographic areas, thereby reducing the influence of broader regional shocks but also absorbing a considerable share of the spatial variation through which the border-access shock operates.

[6]A broader set of robustness exercises is reported in Appendix D. These include alternative geographic samples based on distance-to-border buffers, an absolute-distance treatment measure, nighttime-light levels rather than the logarithmic transformation, municipality-specific linear trends, separate country samples, and an analysis of the subsequent reopening of the border. These exercises generally reinforce the main conclusion that the magnitude and precision of the estimated relationship are sensitive to the treatment definition and to the extent to which local economic trajectories are controlled for.

7.3 Heterogeneity across the border

The pooled estimates may conceal important differences between the Colombian and Venezuelan sides of the border. This possibility is particularly relevant because the same disruption to international connectivity occurred against very different national economic backgrounds. A specification allowing the post-treatment relationship to vary by country therefore provides additional information about the spatial distribution of the estimated effects.

Using municipality and country-by-quarter fixed effects, the estimated coefficient for Colombia is positive, at 0.0128, with a standard error of 0.0053. In contrast, the corresponding coefficient for Venezuela is -0.0509, with a standard error of 0.0177. Both coefficients are statistically different from zero at conventional levels, with the Venezuelan estimate substantially larger in absolute magnitude.

These estimates reveal considerable heterogeneity in the association between changes in border access and nighttime luminosity. Municipalities on the Venezuelan side experiencing larger increases in effective distance to an operational crossing also experienced comparatively weaker nighttime-light outcomes after the disruption. The corresponding conditional relationship for Colombian municipalities is positive.

The difference should nevertheless be interpreted cautiously. Country-specific coefficients do not establish that the border disruption generated opposite causal effects in Colombia and Venezuela, since treatment intensity may remain correlated with pre-existing differences among municipalities within each country. This concern is examined directly using the dynamic specifications below and is particularly important for Venezuela, where the economic crisis generated substantial spatial as well as temporal variation in economic activity. The estimates are therefore most informative as evidence that the relationship between border accessibility and local economic activity was heterogeneous across the two sides of the border.

7.4 Dynamic effects and pre-treatment trends

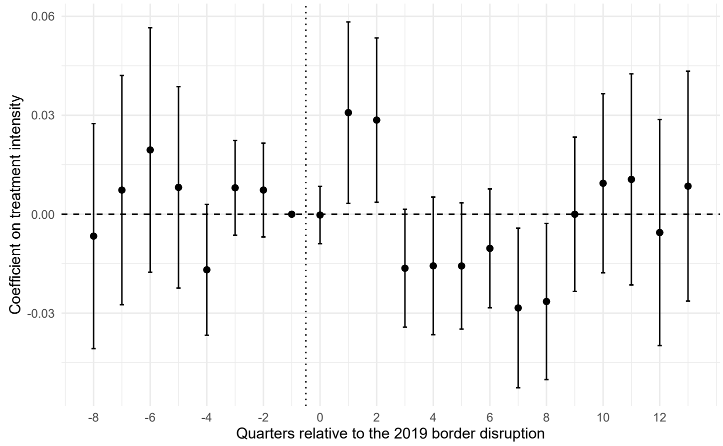
The average post-treatment coefficients summarize the relationship between treatment intensity and nighttime luminosity over the post-2019 period, but they do not reveal how this relationship evolved around the disruption. Event-study specifications are therefore used to examine dynamic patterns before and after 2019. The quarter immediately preceding the disruption, 2018Q4 ($k = -1$), is used as the omitted reference period.

Figure 2 presents the event-study estimates using the most geographically demanding main specification, which combines municipality fixed effects with state/department-by-quarter fixed effects. The coefficients measure the relationship between treatment intensity and nighttime luminosity in each quarter relative to the omitted pre-treatment period.

The dynamic estimates provide a more nuanced picture than the average post-treatment coefficient. Several of the pre-treatment estimates are relatively close to zero and their confidence intervals include zero, particularly in the quarters immediately preceding the disruption. There is also no simple monotonic increase or decrease in the pre-treatment coefficients. Nevertheless, the pre-treatment coefficients are jointly different from zero.

A joint Wald test of the seven pre-treatment coefficients rejects the null that they are jointly equal to zero in the state/department-by-quarter specification ($F = 2.32$, $p = 0.027$). The parallel-trends condition therefore cannot be regarded as fully supported by the data. Importantly, however, the evidence of differential pre-treatment dynamics becomes weaker when the geographic level of the time fixed

Figure 2: Dynamic Border-Access Effects with State/Department-by-Quarter Fixed Effects



Notes: The figure reports event-study coefficients and 95 percent confidence intervals for treatment intensity interacted with quarters relative to the 2019 border disruption. The omitted reference period is 2018Q4 ($k = -1$), while $k = 0$ corresponds to 2019Q1. The specification includes municipality and state/department-by-quarter fixed effects. Standard errors are clustered at the municipality level.

effects becomes more localized. With country-by-quarter fixed effects, the corresponding Wald statistic is $F = 4.64$ ($p < 0.001$), while the crossing-corridor-by-quarter specification produces $F = 4.46$ ($p < 0.001$). The state/department-by-quarter specification reduces the statistic to 2.32, although the null continues to be rejected at the 5 percent level. The complete set of joint pre-treatment tests, including the country-specific specifications, is reported in Appendix C.1.

The post-treatment coefficients also do not display the pattern associated with a simple immediate and persistent treatment effect. The coefficients in the first two quarters following the disruption are positive, after which they become negative for several quarters. The largest negative estimates occur approximately between event times $k = 3$ and $k = 8$, before the coefficients subsequently move back toward zero. Several individual post-treatment estimates are statistically different from zero, but their sign and magnitude vary over time.

This dynamic pattern complicates a simple interpretation in which the 2019 disruption generated an immediate and persistent effect on local economic activity. Rather than displaying a stable post-treatment divergence, the estimated relationship changes sign and magnitude across quarters. Combined with the evidence of differential pre-treatment dynamics, this pattern suggests that changes in border accessibility occurred alongside other spatially heterogeneous economic developments.

As an additional diagnostic, Appendix F examines the subsequent reopening of the international border. Municipalities that had experienced larger increases in effective crossing distance during the closure exhibit comparatively stronger nighttime-light outcomes following the reopening. The estimated reopening coefficient is positive and statistically significant. This reverse-shock pattern is consistent with border accessibility having economic relevance, although it is not treated as an independent causal estimate because crossings reopened at somewhat different dates and the reopening coincided with other changes in economic and bilateral conditions.[7]

[7]The reopening exercise is interpreted as a descriptive validation exercise rather than as an exact inverse treatment of the 2019 disruption. The construction of the reopening period, average estimates, and dynamic results are reported in Appendix F.

8 Discussion and Limitations

The empirical evidence points to an economically meaningful but empirically difficult relationship between border accessibility and local economic activity. The conventional two-way fixed-effects model shows a negative association between deterioration in access to international crossings and nighttime luminosity after 2019. However, this relationship becomes substantially smaller once increasingly localized time-varying shocks are absorbed. The sensitivity of the coefficient indicates that municipalities experiencing different changes in border accessibility were also exposed to different regional economic dynamics.

The event-study results reinforce this interpretation. Differential pre-treatment dynamics remain present even under the most demanding spatial fixed-effects specification, while the post-treatment coefficients vary in sign and magnitude across quarters. These patterns make it difficult to interpret the estimated post-2019 relationship as an isolated causal effect of the border disruption. At the same time, the consistent deterioration in effective network access documented by the GIS analysis, together with the reverse pattern observed around the subsequent reopening, suggests that accessibility remained an economically relevant dimension of the border shock. The evidence is therefore more consistent with border accessibility interacting with broader regional economic conditions than with a single uniform treatment effect.

Three limitations qualify this interpretation. First, the rejection of the parallel-trends assumption prevents a strong causal interpretation of the estimated coefficients. Localized time fixed effects reduce this concern but cannot eliminate unobserved municipality-specific shocks correlated with treatment intensity. Second, nighttime luminosity is a proxy rather than a direct measure of output, employment, or welfare. It provides a consistent measure across both countries, but changes in luminosity may also reflect factors such as electricity provision and infrastructure. Third, the treatment measure captures formal road-network access to operational crossings and therefore does not fully represent informal mobility, differences in road conditions, or the institutional complexity of individual crossing restrictions.

These qualifications imply that the estimates are best interpreted as evidence about the spatial relationship between effective border access and local economic activity rather than as a precise causal estimate of the economic cost of the 2019 disruption. The broader implications of this evidence are developed in the conclusion.

9 Conclusion

This paper examines the relationship between changes in international border accessibility and local economic activity along the Colombia–Venezuela border. The analysis is motivated by the unusual economic geography of the region, where municipalities on both sides of the international boundary have historically depended on cross-border connections for commercial activity. Restrictions on formal crossings in 2019 therefore represented more than a change in bilateral policy: they altered the effective geographic connections between border communities and the markets and economic opportunities located across the boundary.

To measure this change, the paper implements the spatial-accessibility approach developed by Behrens (2024) in the Colombia–Venezuela context. Rather than defining treatment exclusively through proximity to the international boundary, exposure is constructed from changes in shortest-path road-network distance to an operational international crossing. Combining the road network with the location and historical status of formal crossings reveals substantial heterogeneity in the geographic incidence of the border disruption. This spatial measure is combined with quarterly VIIRS nighttime-light observations to examine local economic activity consistently across both countries.

The empirical results initially indicate a negative relationship between the deterioration of border access and nighttime luminosity. This relationship, however, becomes smaller and statistically insignificant when increasingly localized time-varying shocks are absorbed. The attenuation is particularly important once comparisons are restricted to municipalities within the same state or department and quarter.

The dynamic evidence provides an additional qualification. Event-study estimates reveal differential pre-treatment trajectories between municipalities with different levels of subsequent exposure. Also, he joint pre-treatment tests continue to reject the absence of differential trends. The empirical evidence therefore does not support interpreting the baseline coefficient as a clean causal estimate of the economic effect of the 2019 border disruption. Instead, the results indicate that changes in border accessibility occurred alongside important spatial differences in local economic trajectories.

This qualification is itself relevant for the implementation of the spatial research design. The Colombia–Venezuela case demonstrates that a geographically well-defined policy shock does not necessarily provide clean causal identification simply because exposure can be measured precisely. GIS techniques can substantially improve treatment measurement by reconstructing how a policy change modifies effective economic geography, but they cannot by themselves eliminate differences in the trajectories of locations receiving different levels of exposure. The distinction between treatment measurement and treatment identification is therefore central to the interpretation of the results.

Nevertheless, the analysis provides evidence that border accessibility is an economically meaningful dimension of the Colombia–Venezuela border economy. The network calculations show that changes in crossing availability generated large increases in effective travel distance for many municipalities despite leaving their physical distance to the international boundary unchanged. The pooled estimates retain a negative sign across the principal specifications, although their magnitude and precision decline under more demanding controls. In addition, the reopening analysis produces a reverse pattern in which municipalities more exposed to the earlier deterioration in accessibility subsequently experience comparatively stronger nighttime-light outcomes. These findings are consistent with an economic role for border access, although they do not overcome the identification concerns present in the main analysis.

Future research could strengthen the identification of these effects by combining the spatial framework developed here with more localized information on trade, employment, firm activity, migration, or cross-border mobility. More detailed information on traffic volumes and the use of formal and informal crossings could also improve the measurement of effective accessibility beyond the operational status of individual crossings.

The central conclusion is therefore not that the 2019 disruption can be assigned a single precisely identified local economic effect. Rather, the evidence shows that border policy substantially altered the effective economic geography of the Colombia–Venezuela frontier and that these changes were systematically related to the evolution of local economic activity. In a setting where conventional economic information is limited, the combination of GIS, transportation networks, and satellite observations provides a useful framework for making these spatial relationships visible while remaining explicit about the limits of causal inference.

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A International crossing regimes

The construction of the accessibility measure requires identifying which formal international crossings were operational under each border regime. Table 2 summarizes the classification used in the empirical analysis.

The relevant concept of “open” refers to formal vehicular, cargo, or foreign-trade accessibility rather than pedestrian passage or informal cross-border movement. This distinction is important because the treatment measure is based on road-network accessibility to formal international crossings.

The institutional evidence provides additional support for the treatment construction. The rerouting implied by the network model is not merely a geographic assumption. MINCIT explicitly reports that, following the February 2019 closure of cargo access through Norte de Santander, merchandise continued through Paraguachón and Cartagena. The Ministry also documents the longer Cartagena–Maicao–Guarero–Ureña route and the associated increase in transportation costs (Ministerio de Comercio, Industria y Turismo, 2022b). The network-based treatment measure therefore captures a documented change in formal trade geography.

Table 2: Operational status of international border crossings

Crossing	Before Feb. 2019	Feb. 2019–Sep. 2022	Reopening	Institutional evidence
Simón Bolívar (C01)	Open	Closed	26 Sep. 2022	MINCIT identifies February 2019 as the final closure of cargo access through Norte de Santander and documents the restoration of cargo traffic through Simón Bolívar on 26 September 2022 (Ministerio de Comercio, Industria y Turismo, 2022b,a).
Francisco de Paula Santander (C02)	Open	Closed	26 Sep. 2022	MINCIT identifies Francisco de Paula Santander together with Simón Bolívar as the two bridges reopened for cargo traffic on 26 September 2022 (Ministerio de Comercio, Industria y Turismo, 2022b, 2023).
Atanasio Girardot (C03)	Not operational	Not operational	1 Jan. 2023	The Ministry of Transport records the official entry into operation of Atanasio Girardot on 1 January 2023. MINCIT independently distinguishes this opening from the September 2022 reopening of C01 and C02 (Ministerio de Transporte, 2023; Ministerio de Comercio, Industria y Turismo, 2023).
Paraguachón (C04)	Open	Open	Remained open	MINCIT states that Paraguachón remained open following the February 2019 closure of the Norte de Santander and Arauca cargo crossings and that merchandise was redirected through Paraguachón and Cartagena. DIAN classifies Paraguachón as an official terrestrial border crossing (Ministerio de Comercio, Industria y Turismo, 2022a,b; Dirección de Impuestos y Aduanas Nacionales, 2026).
José Antonio Páez (C05)	Qualified[a]	Closed	Jan. 2023	Official Colombian documentation identifies the Arauca crossing as closed from February 2019, while the Ministry of Transport reports the normalization of the Arauca border crossings from January 2023 (Congreso de Colombia, 2023; Ministerio de Transporte, 2023).

Notes: "Open" refers to formal vehicular, cargo, or foreign-trade accessibility rather than pedestrian or informal movement. The February 2019–September 2022 period constitutes the principal closure regime used in the treatment construction.
[a] The pre-2019 cargo status of José Antonio Páez is less clearly documented than that of C01, C02, and C04 and is therefore not treated as a central source of identifying variation.

B Descriptive statistics

Table 3: Descriptive Statistics: Baseline Estimation Sample

Variable	N	Mean	SD	Median	Min.	Max.
Nighttime lights	4,032	0.727	2.850	0.115	0.000	41.900
Log nighttime lights	4,032	0.294	0.498	0.109	0.000	3.760
Pre-treatment network distance (km)	192	168.0	92.9	168.0	4.96	424.0
Post-treatment network distance (km)	192	513.0	201.0	530.0	13.7	993.0
Change in network distance (km)	192	345.0	210.0	434.0	0.0	737.0
Treatment intensity	192	1.20	0.897	1.20	0.00	4.60
Distance to international border (km)	192	57.0	33.5	57.0	1.97	133.0

Notes: Nighttime-light variables vary by municipality and quarter and therefore contain 4,032 observations. Geographic and treatment variables are time invariant at the municipality level and consequently contain 192 observations. Treatment intensity is the difference between the logarithm of post-treatment and pre-treatment shortest road-network distance to the nearest operational international crossing.

C Additional identification diagnostics

C.1 Joint pre-treatment tests

The event-study specifications reported in the main text are complemented by joint tests of the pre-treatment coefficients. The null hypothesis is that the seven pre-treatment coefficients are jointly equal to zero. As discussed in Section 7.4, evidence of differential pre-treatment dynamics becomes weaker under increasingly localized time fixed effects but does not disappear completely.

Table 4: Joint Tests of Pre-Treatment Coefficients

Specification	F statistic	p-value	df_1	df_2
Country × Quarter FE	4.64	< 0.0001	7	191
Crossing Corridor × Quarter FE	4.46	0.0001	7	191
State/Department × Quarter FE	2.32	0.0272	7	191
Colombia	3.74	0.0010	7	131
Venezuela	6.32	< 0.0001	7	59

Notes: Joint Wald tests of the null that the seven pre-treatment event-study coefficients are jointly equal to zero. Standard errors are clustered at the municipality level. The omitted event-study period is 2018Q4 ($k = -1$).

D Robustness to alternative samples and specifications

D.1 Alternative border buffers

The sensitivity of the estimates to the geographic extent of the sample is evaluated using alternative restrictions on municipality distance from the international boundary.

The smallest 50 km sample produces a positive and statistically imprecise estimate, while the 100 km and broader samples produce negative estimates close to the preferred state/department-by-quarter specification. The main interpretation is therefore not driven by a single geographic cutoff.

D.2 Alternative treatment, outcome, and trend specifications

Table 5: Robustness to Alternative Border Buffers

Buffer (km)	Municipalities	Observations	Estimate	Std. Error	p-value
50	87	1,827	0.0064	0.0136	0.638
100	169	3,549	-0.0051	0.0121	0.675
150	192	4,032	-0.0057	0.0114	0.616
300	192	4,032	-0.0057	0.0114	0.616

Notes: Each row reports the coefficient on treatment intensity interacted with the post-2019 indicator for the indicated maximum distance from the international boundary. Specifications include municipality and state/department-by-quarter fixed effects. Standard errors are clustered at the municipality level.

Table 6 reports additional specifications that modify the treatment definition, dependent variable, and treatment of local economic trends.

Table 6: Alternative Specifications

	(1) Reference	(2) Absolute distance	(3) NTL levels	(4) Municipality trends
Treatment intensity × Post	−0.006		−0.088	−0.000
	(0.011)		(0.069)	(0.015)
Distance increase (100 km) × Post		−0.006		
		(0.005)		
Outcome	Log NTL	Log NTL	NTL levels	Log NTL
Municipality FE	Yes	Yes	Yes	Yes
State/Department × Quarter FE	Yes	Yes	Yes	Yes
Municipality-specific linear trends	No	No	No	Yes
Observations	4,032	4,032	4,032	4,032
R^2	0.986	0.986	0.948	0.989
Adjusted R^2	0.984	0.984	0.941	0.987
Within R^2	0.001	0.001	0.001	0.000
Adjusted within R^2	0.000	0.001	0.001	-0.000

Notes: Columns (1), (3), and (4) use treatment intensity $\log(d_i^{post}) - \log(d_i^{pre})$. Column (2) instead uses the absolute increase in road-network distance measured in 100-km units. The dependent variable is $\log(1 + \text{nighttime lights})$ except in Column (3), which uses nighttime-light levels. All specifications include municipality and state/department-by-quarter fixed effects. Column (4) additionally allows each municipality to follow a separate linear time trend. Standard errors are clustered at the municipality level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Using the absolute increase in road-network distance instead of the logarithmic treatment measure does not produce a statistically significant relationship. The same is true when nighttime-light levels are used directly. Allowing municipalities to follow separate linear trends further reduces the estimated average post-treatment relationship. These exercises reinforce the sensitivity of the estimated magnitude to functional-form assumptions and local economic dynamics.

E Country-specific evidence

The pooled analysis masks substantial heterogeneity between the Colombian and Venezuelan sides of the border. Table 7 reports country-specific estimates, while Figure 3 presents the corresponding dynamic specifications.

The country-specific estimates indicate substantially different conditional relationships across the two sides of the border. However, the corresponding event studies also reveal differential pre-treatment

Table 7: Country-Specific Estimates

	(1) Joint model	(2) Colombia	(3) Venezuela
Colombia treatment intensity × Post	0.013^{**}	0.020^{*}	
	(0.005)	(0.010)	
Venezuela treatment intensity × Post	-0.051^{***}		-0.037
	(0.018)		(0.028)
Municipality FE	Yes	Yes	Yes
Country × Quarter FE	Yes	No	No
State/Department × Quarter FE	No	Yes	Yes
Observations	4,032	2,772	1,260
R^2	0.984	0.983	0.987
Adjusted R^2	0.983	0.980	0.985
Within R^2	0.045	0.007	0.020
Adjusted within R^2	0.045	0.007	0.019

Notes: Column (1) estimates Colombia- and Venezuela-specific treatment coefficients jointly using municipality and country-by-quarter fixed effects. Columns (2) and (3) estimate the model separately for Colombian and Venezuelan municipalities using municipality and state/department-by-quarter fixed effects. Treatment intensity is $\log(d_i^{post}) - \log(d_i^{pre})$. The dependent variable is $\log(1 + \text{nighttime lights})$. Standard errors are clustered at the municipality level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

dynamics. The estimates are therefore interpreted as evidence of heterogeneity rather than clean country-specific causal effects.

F Border reopening

The subsequent restoration of formal border connectivity provides a complementary reverse-shock diagnostic. The exercise examines municipalities that experienced larger deteriorations in accessibility during the closure subsequently experienced comparatively stronger nighttime-light outcomes as formal access was restored.

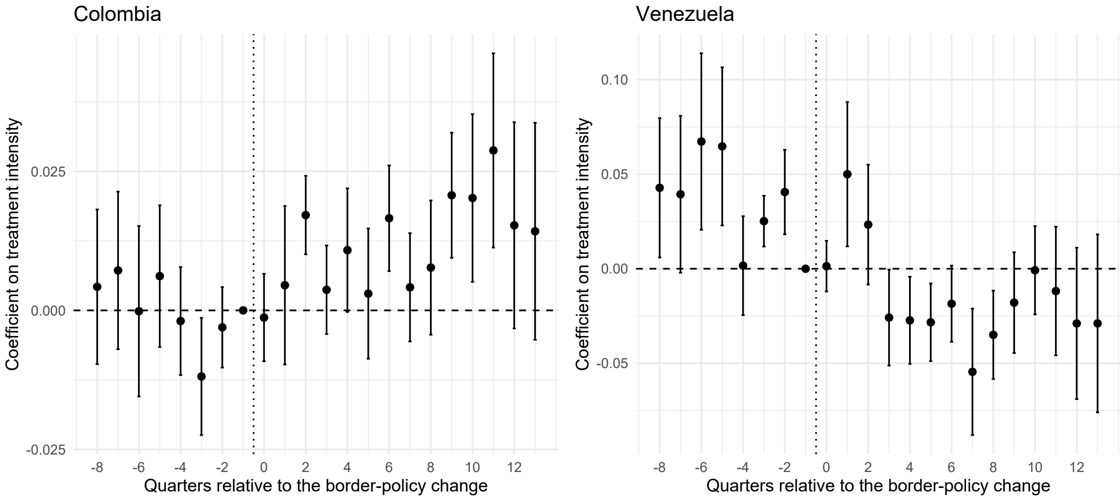


Figure 3: Country-specific event-study estimates

Notes: The figure reports event-study estimates separately for Colombian and Venezuelan municipalities. The omitted reference period is 2018Q4 ($k = -1$). Standard errors are clustered at the municipality level. Country-specific pre-treatment coefficients are jointly different from zero, so the dynamic estimates are interpreted as descriptive evidence of heterogeneity rather than separate causal effects.

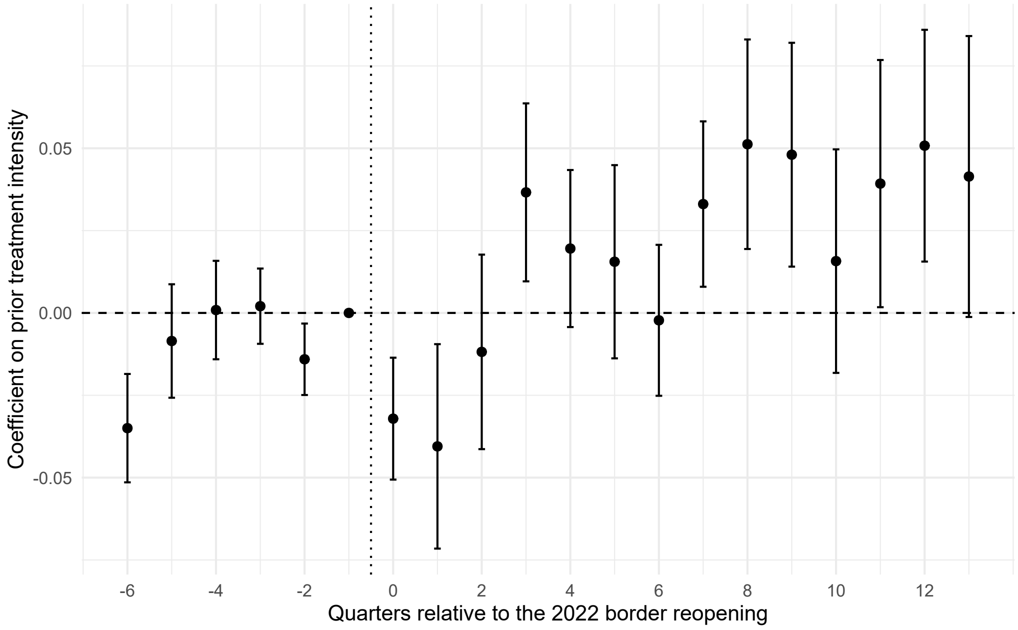
Table 8: Border-Reopening Diagnostic

	(1) Reopening diagnostic
Prior exposure × Reopening	0.032**
	(0.012)
Municipality FE	Yes
State/Department × Quarter FE	Yes
Observations	3,648
R^2	0.988
Adjusted R^2	0.986
Within R^2	0.015
Adjusted within R^2	0.014

Notes: The dependent variable is log(1 + nighttime lights). Prior exposure corresponds to the treatment intensity generated by the 2019 closure regime. The specification includes municipality and state/department-by-quarter fixed effects. Standard errors are clustered at the municipality level. The reopening exercise is interpreted as a descriptive reverse-shock diagnostic rather than as an independent causal estimate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The average reopening coefficient is positive and statistically significant. This reverse pattern is consistent with an economic role for border accessibility, but it does not resolve the identification concerns documented in the main analysis.

Figure 4: Dynamic evidence around the border reopening



Notes: The figure reports the interaction between prior treatment intensity and quarters relative to the reopening period. The reopening exercise is interpreted as a descriptive validation exercise rather than as an exact inverse treatment of the 2019 disruption because the restoration of formal access coincided with other political and economic changes.